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SUMMARY

Machine Learning Engineer with 5+ years of experience across industry and academic research, specializing in computer vision, multimodal deep learning, and LLM systems. Designs, trains, and deploys production-grade models spanning object detection, pose estimation, Retrieval-Augmented Generation (RAG), and edge-deployed vision, turning applied research into real-world AI products. Author of peer-reviewed publications at IEEE ICASSP and SCIA in 3D pose estimation and physiological signal analysis.

SKILLS

- **Languages & Frameworks:** Python, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
- **Computer Vision:** Object Detection & Tracking (YOLO, Mask R-CNN, DeepSORT, ByteTrack), Segmentation, Pose Estimation (OpenPose, ViTPose), Facial Recognition
- **NLP & LLMs:** LLMs, Retrieval-Augmented Generation (RAG), Multimodal Fusion, Image Captioning
- **Signal & Sensing:** Physiological & Motion Signal Extraction, Multimodal Sensor Fusion, Signal Processing
- **MLOps & Deployment:** Docker, Flask, AWS (S3, EC2, SageMaker), GCP, Git, MySQL, MongoDB, Linux
- **Edge AI:** NVIDIA Jetson, Coral TPU, Hailo, Raspberry Pi

Tharindu Ekanayake

EXPERIENCE

January 2026 - Present

Machine Learning Engineer, goAVA GmbH | Essen, Germany

- Build and deploy production machine learning models and Retrieval-Augmented Generation (RAG) systems, integrating Large Language Models (LLMs) with custom retrieval pipelines to deliver context-aware, knowledge-grounded AI solutions.

October 2024 - January 2026

Project Researcher / Research Assistant Center for Machine Vision and Signal Analysis, University of Oulu | Oulu, Finland

- Designed and trained AI models to extract physiological signals (heart rate, respiration, posture, and gaze) from contactless sensors, video, audio, and thermal imaging within immersive XR environments.
- Built multimodal data-integration and real-time processing pipelines that fuse inputs from cameras, microphones, and thermal sensors to adapt XR experiences to the user's physiological and cognitive state.
- Contributed to the AugmentCare project, applying segmentation, pose estimation, and signal processing to model cognitive load and stress responses in virtual reality (VR), using both supervised and unsupervised approaches for accurate user-state classification.

October 2020 - January 2024

Machine Learning Engineer, ZOOMi Technologies Inc. | Colombo, Sri Lanka

- Delivered end-to-end machine learning solutions across a diverse portfolio spanning the medical sector, autonomous vehicle navigation for agriculture, and AI-powered self-checkout retail systems.
- Designed and engineered computer vision models for object detection, tracking, segmentation, classification, and pose estimation using PyTorch, TensorFlow, YOLO, Mask-RCNN, DeepSORT, and ByteTrack.
- Built and deployed Large Language Model (LLM) applications with Retrieval-Augmented Generation (RAG) pipelines for automation and information-retrieval use cases.
- Played a key role in developing LayerNext, a computer vision data-management and model-training platform, and optimized models for edge deployment on NVIDIA Jetson and Coral TPU devices.

EDUCATION

May 2025

Master of Science | Computer Science and Engineering, AI University of Oulu, Oulu, Finland

April 2020

Bachelor of Science | Electrical and Electronic Engineering University of Peradeniya, Kandy, Sri Lanka

PUBLICATIONS

3DPCNet: Pose Canonicalization for Robust Viewpoint-Invariant 3D Kinematic Analysis from Monocular RGB Cameras

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026

- A pose-canonicalization module that maps any input 3D pose into a consistent body-centered frame for viewpoint-invariant kinematic analysis from monocular RGB video.

Evaluating the Accuracy and Reliability of Camera-Based Physiological and Motion Signal Extraction Techniques in Virtual Reality Training Environments

Scandinavian Conference on Image Analysis (SCIA), Springer LNCS, 2025

- A systematic evaluation of contactless, camera-based methods for extracting physiological and motion signals from users during VR training.

Unsupervised Multimodal Biosignal Analysis of Student Cognitive and Physiological Responses During VR-Based Training Sessions

M.Sc. Thesis, University of Oulu, 2025

- Unsupervised, multimodal modeling of students' cognitive load and engagement from biosignals recorded during VR-based training.